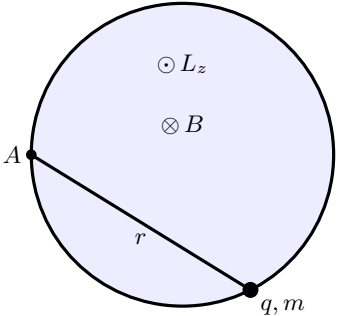
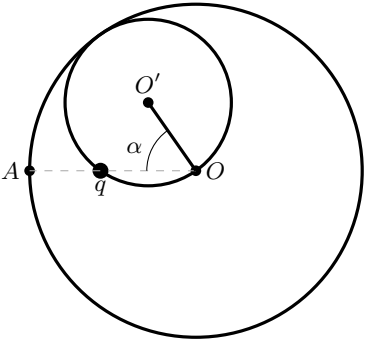


$L_z - \frac{1}{2}qBr^2 =$ 常量

磁场不做功



图取 $\alpha = 55°$；小环确在大环内

小珠在 AO 上：$r = R(1 - \cos\alpha)$

$\beta = 2\alpha$ 是相对接触半径的夹角